AETHERIX

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

Building an Interplanetary Communication Network with DTN, Quantum Communication, and Space-Based Infrastructure for Mars Mission Support

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matx104.github.io/AETHERIX | github.com/matx104/AETHERIX

PRESENTATION AGENDA

01 The Challenge
Why interplanetary communication is hard

02 AETHERIX Architecture
DTN + AI Routing + Quantum Security

03 DTN & Bundle Protocol v7
Store-and-forward networking foundation

04 5-Tier Network Topology
241 nodes across Earth and Mars

05 Optical Link Budget
1550 nm laser performance analysis

06 RL-Based Routing
Multi-agent federated Q-learning

07 Quantum Security (QKD)
BB84/E91 with repeater chains

08 Orbital Mechanics
Contact windows and synodic period

09 Mars Mission Scenario
End-to-end simulation walkthrough

10 Performance Comparison
AETHERIX vs current systems

11 Roadmap & Standards
CCSDS, IETF, deployment phases

12 Conclusion & Q&A
Summary and live demo

[A2] AETHERIX topology.py (241 nodes) · References slide at end

WHAT IS AETHERIX?

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

What is AETHERIX?

An architecture for delay-tolerant networking (DTN) between Earth and Mars, combining:

- Bundle Protocol v7 (BPv7) via ION-DTN
- Reinforcement Learning routing (replacing static CGR)
- Quantum Key Distribution (QKD) for security
- Hybrid optical/RF communications
- 5-tier topology with 241 nodes

The Problem

- Earth-Mars: 54.6M – 401M km distance [3]
- One-way light delay: 3 – 22 minutes [3]
- Solar conjunction: 2-week total blackout
- TCP/IP fundamentally cannot work [12]
- Current RF: 0.5 – 6 Mbps (NASA MRO) [1]

AETHERIX delivers a complete interplanetary networking solution

combining DTN protocols, AI-driven routing, quantum-secure keys, and hybrid optical/RF links for Mars mission support.

10-100×	>95%	241	480
Target	Availability (target)	Nodes [A2]	Tests

[1] NASA MRN 2024 · [3] JPL Horizons · [12] RFC 4838 · [A2] topology.py

THE DISTANCE

Why Space Communication is Fundamentally Different

Earth → Mars	Distance	Light Time	Data Rate
Perihelion (closest)	54.6M km	~3 min	100-200 Mbps
Average	225M km	~12.5 min	10-20 Mbps
Aphelion (farthest)	401M km	~22 min	2-5 Mbps

TCP/IP Assumption	Space Reality	Impact
RTT < 1 second	6 – 44 min RTT [3]	Timeout failure
Always connected	Scheduled contacts	Connection drops
End-to-end path	No persistent path	Routing impossible
Fast acknowledgments	ACKs take minutes [3]	Window collapse
Low packet loss	High loss (optical)	Retransmit sto

"At aphelion, a single ping-pong takes 44 minutes.
No TCP session can survive that. We need an entirely
different networking paradigm."

[3] JPL Horizons (distance/light-time) · [12] RFC 4838 (DTN rationale) · [1] NASA MRO data rate

THE ANSWER: DELAY-TOLERANT NETWORKING

Decouple Communication from Connectivity

CREATE
Bundle

STORE
Locally

WAIT
For Link

FORWARD
Next Hop

DELIVER
Destination

BPv7

RFC 9171

Store-and-forward
Custody transfer

AI Routing

RL Agent

Multi-agent federated
Q-learning

QKD

BB84/E91

Quantum key dist.
Future-proof security

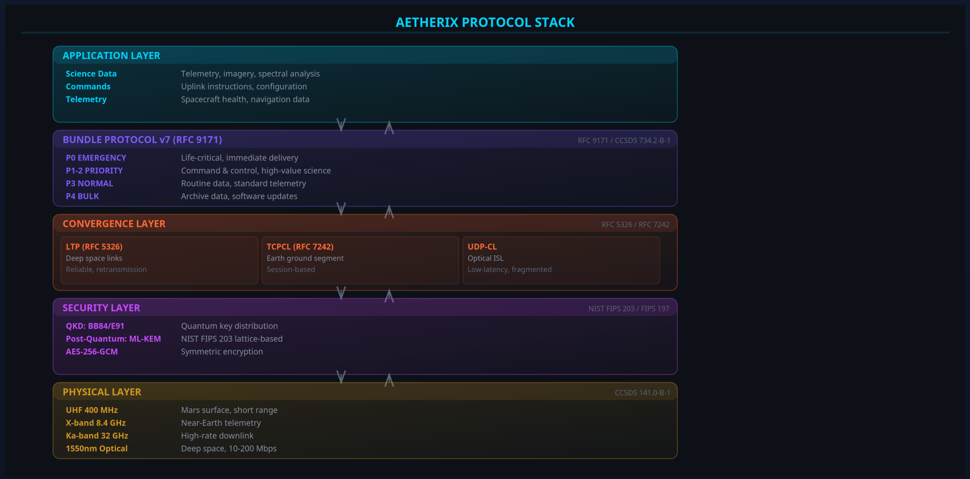
Key Insight: DTN moves reliability from end-to-end to hop-by-hop. Each custodian is responsible until the next node accepts custody — enabling communication even with 44-minute round-trip delays.

[9] RFC 9171 BPv7 · [10] RFC 5326 LTP · [11] RFC 7242 TCPCL · [12] RFC 4838

SYSTEM ARCHITECTURE

5 Integrated Modules for Interplanetary Communication

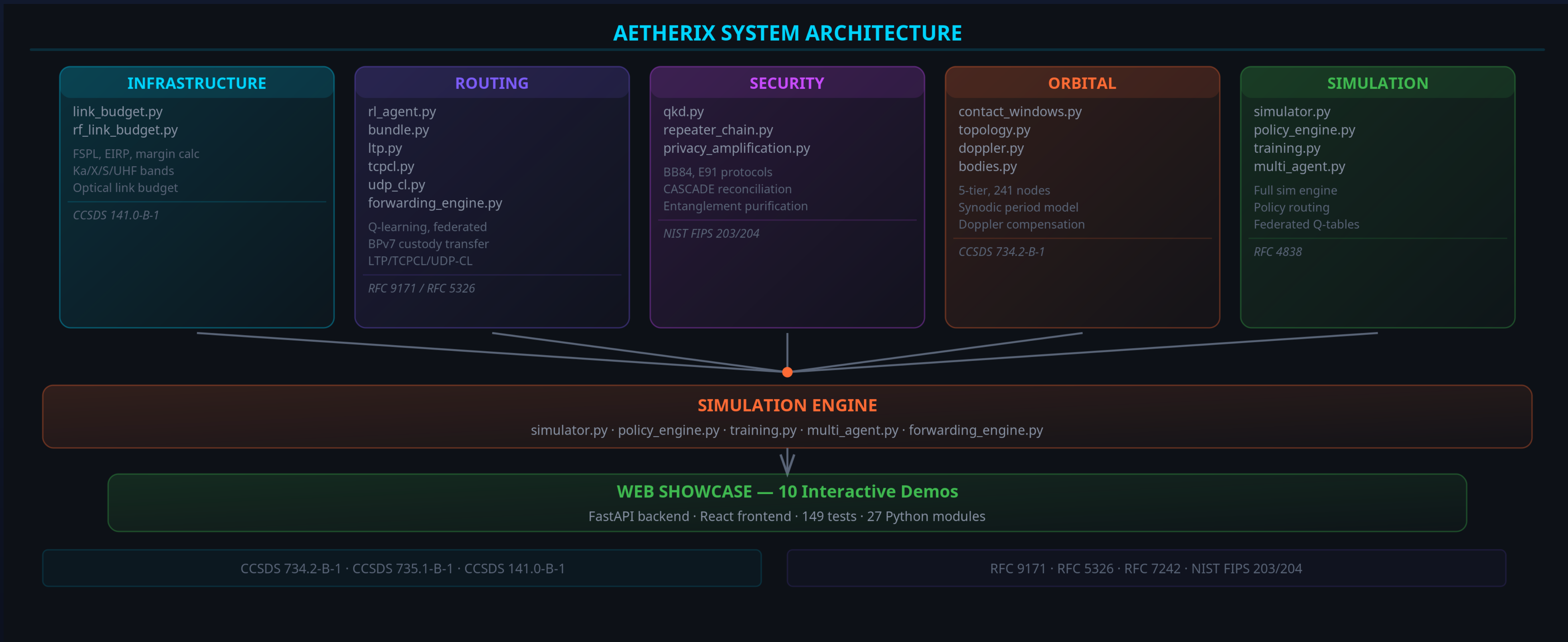
Infrastructure	Link budget calculations, optical/RF analysis
Routing	RL agents, BPv7 bundles, forwarding engine
Security	BB84/E91 QKD, repeater chains, privacy amplification
Orbital	Contact windows, Doppler, celestial body database
Simulation	End-to-end scenario engine, policy routing



Layer	Module	Engine	Showcase
Infrastructure	link_budget.py	OpticalLinkBudget	Earth-Mars scenarios
Routing	rl_agent.py	RLRoutingAgent	Federated Q-learning
Security	qkd.py	BB84/E91	Quantum key exchange
Orbital	contact_windows.py	ContactPredictor	Synodic period
Simulation	simulator.py	SimEngine	Full mission sim

[A1] rl_agent.py · [A2] topology.py · [A3] simulator.py · [A4] link_budget.py · [A5] qkd.py

SYSTEM ARCHITECTURE DIAGRAM



BUNDLE PROTOCOL v7 — DEEP DIVE

RFC 9171 — The Foundation of Interplanetary Networking

APPLICATION LAYER

Science data, commands, imagery

BUNDLE PROTOCOL v7 (RFC 9171)

Store-and-Forward | Custody Transfer | Priority (P0–P4)

CONVERGENCE LAYERS

LTP (deep space) | TCPCL (Earth) | UDP-CL (optical ISL)

PHYSICAL LAYER

Optical 1550nm (2–200 Mbps) | RF Ka-band | UHF (surface)

Priority	Name	Delay Target	Example
P0	Emergency	< 1 min	Anomaly alerts
P1	Expedited	< 30 min	Solar storm warnings
P2	Standard	< 24 hrs	Science data
P3	Normal	< 7 days	Telemetry
P4	Bulk	< 30 days	Software updates

HOW IT WORKS

1. Source creates bundle (payload + metadata + lifetime)
2. Forward to next hop when link becomes available
3. Store locally during outages and disruptions
4. Custody transfer — next node accepts responsibility
5. Repeat hop-by-hop until destination reached

STANDARDS COMPLIANCE

RFC 9171 — Bundle Protocol Version 7 [9]

RFC 4838 — DTN Architecture [12]

RFC 5326 — Licklider Transmission Protocol [10]

RFC 9172 — Bundle Protocol Security (BPSec)

CCSDS 734.2-B-1 — CCSDS Bundle Protocol [2]

CCSDS 734.3-B-1 — Schedule-Aware Bundle Routing

[9] RFC 9171 · [2] CCSDS 734.2-B-1 · [10] RFC 5326 LTP · [11] RFC 7242 TCPCL · [12] RFC 4838

5-TIER NETWORK TOPOLOGY

241 Nodes — No Single Point of Failure

TIER 1: Earth Ground (6 nodes)

DSN Goldstone, Madrid, Canberra + Mission Control

TIER 2: Earth Orbital (51 nodes)

3 GEO relays + 48 LEO laser constellation

TIER 3: Deep Space (4 nodes)

ES-L4, ES-L5 Lagrange relays + Transfer orbit

TIER 4: Mars Orbital (4 nodes)

2 Areostationary + 2 polar orbiters

TIER 5: Mars Surface (176 nodes)

Bases, rovers, drones, sensor network

DESIGN PHILOSOPHY

Multiple redundant paths at every tier ensure no single point of failure. Lagrange point relays provide coverage during solar conjunction blackouts. Quantum repeaters co-located at L4/L5 enable end-to-end key distribution.

[A2] AETHERIX topology.py (241 nodes, 5 tiers) · [1] NASA Deep Space Network (Goldstone/Madrid/Canberra)

5-TIER NETWORK DETAIL

Tier Breakdown & Link Characteristics

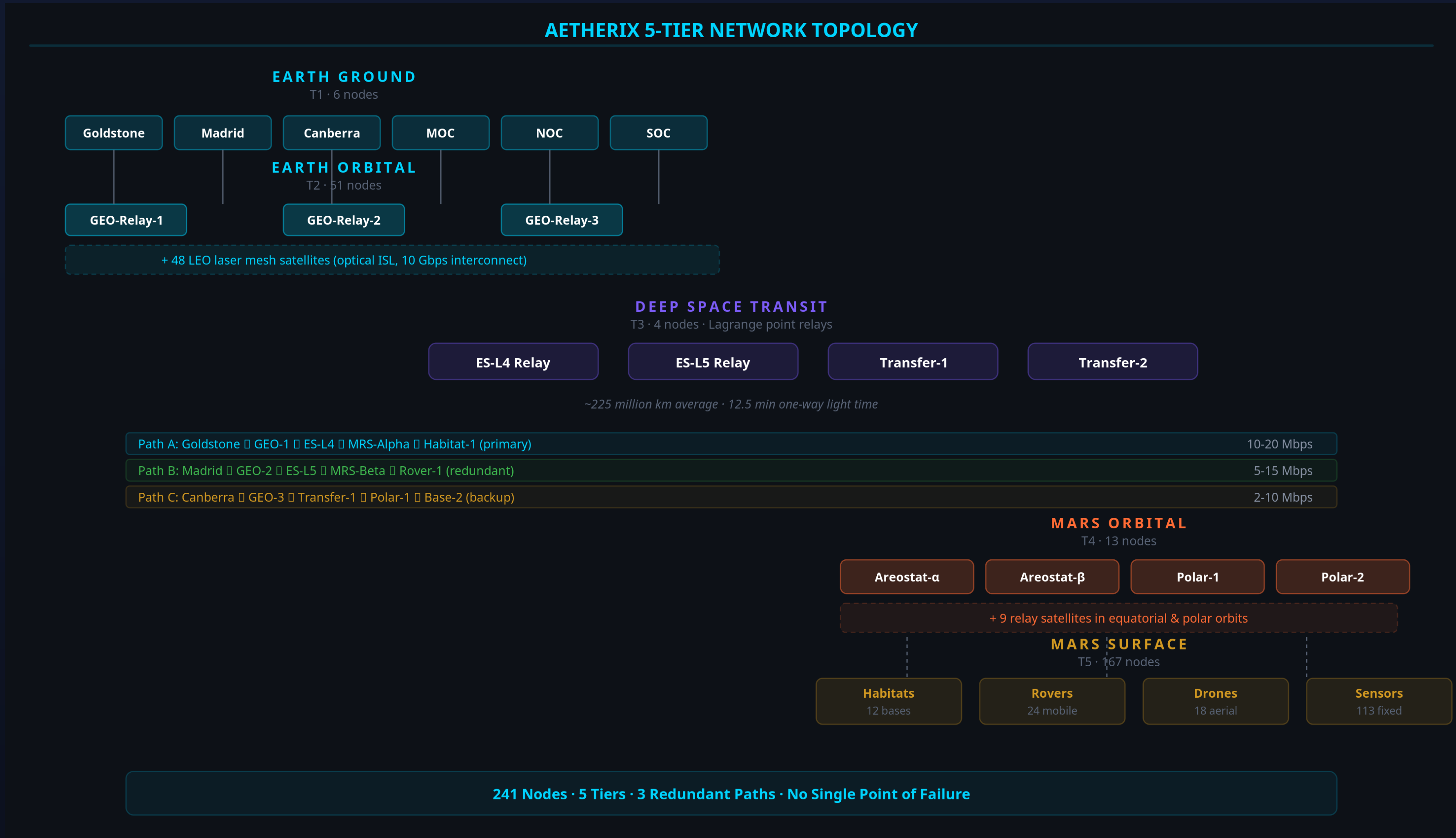
Tier	Name	Nodes	Description
T1	Earth Ground	6	DSN: Goldstone, Madrid, Canberra + MOC, NOC, SOC
T2	Earth Orbital	51	3 GEO relays + 48 LEO laser mesh constellation
T3	Deep Space	4	ES-L4 & ES-L5 Lagrange relays + 2 transfer orbit sats
T4	Mars Orbital	13	2 areostationary + 2 polar orbiters + 9 relay sats
T5	Mars Surface	167	Habitats, rovers, drones, sensor networks

Link	Data Rate	Type
Earth ↔ Deep Space	100 Mbps	1550nm optical
Deep Space ↔ Mars	2–200 Mbps	Distance dependent
Mars Orb ↔ Surface	2 Mbps	UHF/X-band
LEO ISL	10 Gbps	Optical inter-satellite

3 redundant paths · No single point of failure · Lagrange relays for conjunction coverage

5-TIER NETWORK DIAGRAM

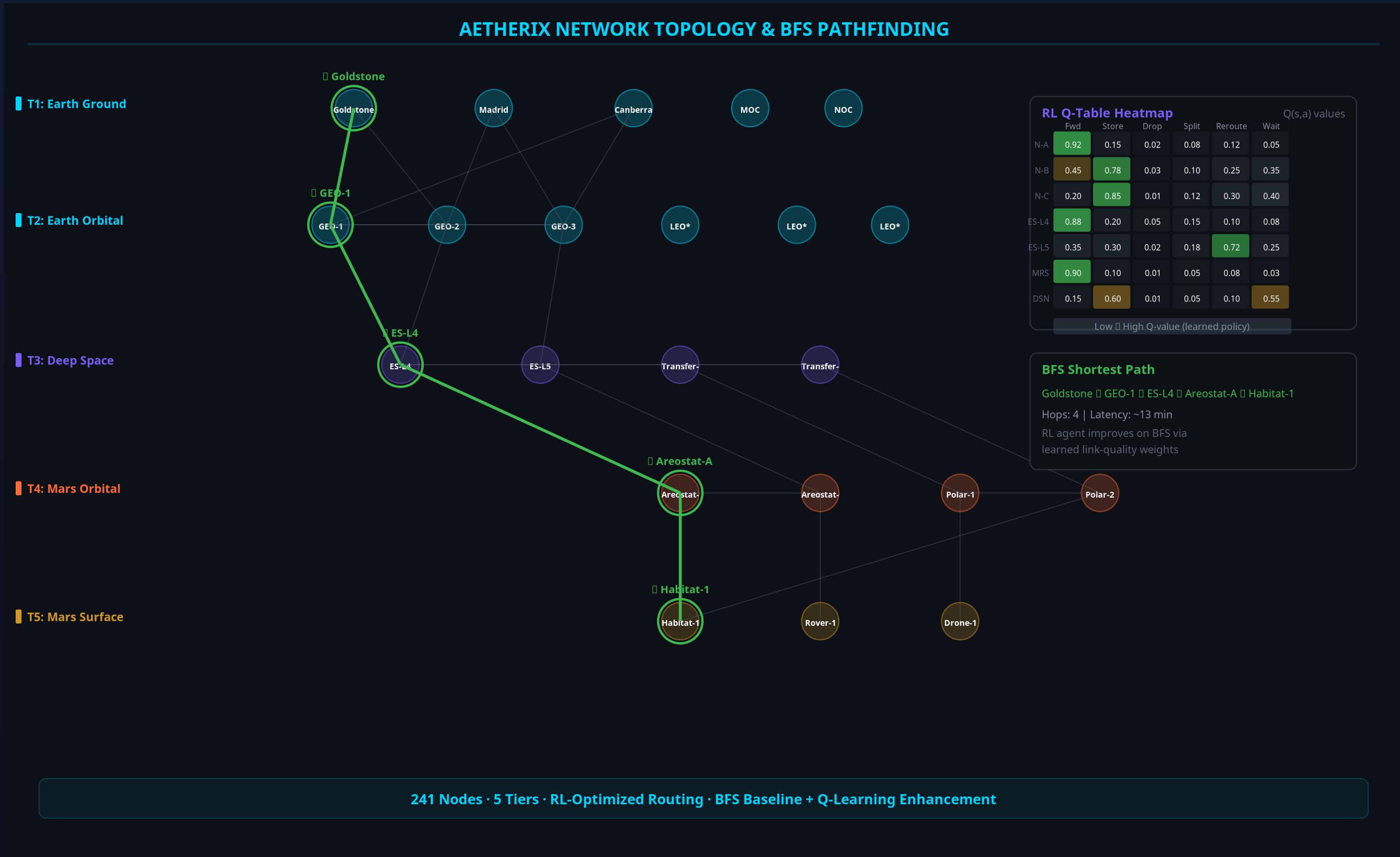
241 Nodes from Earth to Mars Surface



[A2] AETHERIX topology.py · [3] JPL Horizons (Lagrange ES-L4/L5 geometry)

NETWORK TOPOLOGY GRAPH

241 Nodes — BFS Pathfinding + RL-Optimized Routing



	Fwd	Store	Drop	Split	Reroute	Wait
N-A	0.92	0.15	0.02	0.08	0.12	0.05
N-B	0.45	0.78	0.03	0.10	0.25	0.35
N-C	0.20	0.85	0.01	0.12	0.30	0.40
ES-L4	0.88	0.20	0.05	0.15	0.10	0.08
ES-L5	0.35	0.30	0.02	0.18	0.72	0.25
MRS	0.90	0.10	0.01	0.05	0.08	0.03
DSN	0.15	0.60	0.01	0.05	0.10	0.55

OPTICAL COMMUNICATIONS

1550 nm Laser — 10–100× Capability Over RF

200 Mbps
Peak Data Rate
At perihelion (54.6M km)

1550 nm
Wavelength
Telecom heritage lasers

33×
vs Current RF
MRO peak: 6 Mbps

KEY EQUATIONS (CCSDS 141.0-B-1 [5])

$FSPL = (4\pi d/\lambda)^2$ Free-Space Path Loss [5]

$G = (\pi D/\lambda)^2$ Antenna Gain

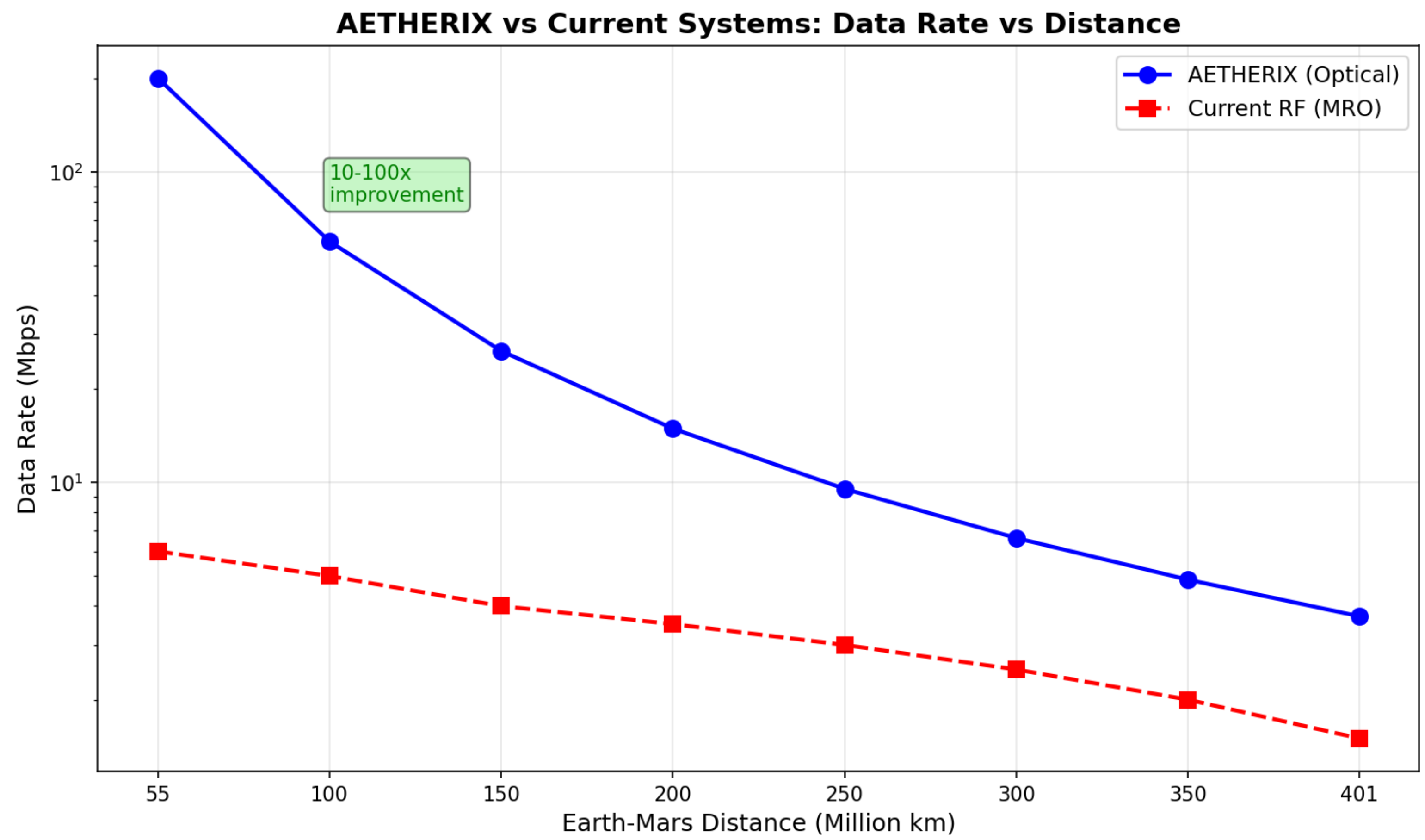
$Pr = Pt \cdot Gt \cdot Gr / FSPL$ Received Power [A4]

Parameter	Value	Rationale
TX Power	5 W (37 dBm)	Space-qualified laser
TX Aperture	22 cm	Mars orbiter class
RX Aperture	1.0 m	DSN-scale telescope
Wavelength	1550 nm	Telecom heritage
Efficiency	0.55	Conservative estimate

[5] CCSDS 141.0-B-1 (optical link) · [A4] AETHERIX link_budget.py · [1] NASA MRO (6 Mbps RF baseline)

DATA RATE VS DISTANCE

1550 nm Optical Link Performance



200 Mbps at closest approach to 2 Mbps at maximum distance · 10–100× capability over RF [1]

EARTH–MARS DATA JOURNEY

7 Hops from Perseverance Rover to Mission Control

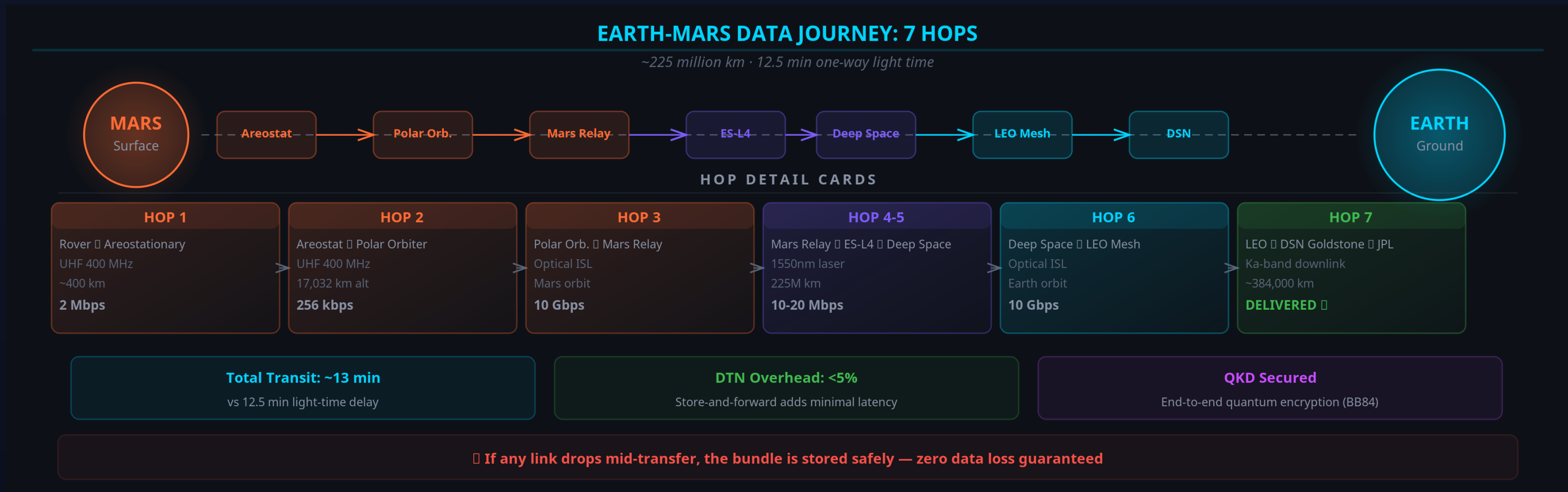
Hop	From	To	Link	Time
1	Perseverance	MRS-Alpha	UHF	~5s
2	MRS-Alpha	MRS-Polar	Optical ISL	~1s
3	MRS-Polar	ES-L5	1550nm laser	~12.5 min
4	ES-L5	ES-L4	Optical ISL	~2s
5	ES-L4	LEO Constellation	1550nm laser	~0.1s
6	LEO Constellation	DSN Madrid	Fiber downlink	~1s
7	DSN Madrid	JPL MOC	Terrestrial fiber	~0.5s

~13 min
Total Transfer [3]

98.7%
Delivery (target)

15 Mbps
Throughput (target)

7 hops
Path Length [A2]



[A2] topology.py (7-hop path) · [3] JPL Horizons (light-time) · delivery/throughput are design targets

RL-BASED ROUTING — AI INNOVATION

Multi-Agent Federated Q-Learning Replaces Static CGR

CGR LIMITATIONS

- Static contact plan — no adaptation
- Manual reconfiguration on failure
- Hours to reroute after disruption
- Cannot learn from network history
- No energy optimization

RL AGENT ADVANTAGES

- Adaptive routing from network state
- 4 actions: FORWARD, STORE, DROP, SPLIT
- $R = \alpha \cdot del - \beta \cdot delay - \gamma \cdot hops - \delta \cdot drop - \varepsilon \cdot energy$ [A1]
- Weights: $\alpha=1.0$, $\delta=10.0$, ε-greedy ε-decay 0.995 [A1]
- Federated learning across agents

713/800

Forward Decisions [A3]

Training run (Module 3)

seconds

Recovery vs hours

Qualitative: auto vs manual [A1]

Q-table

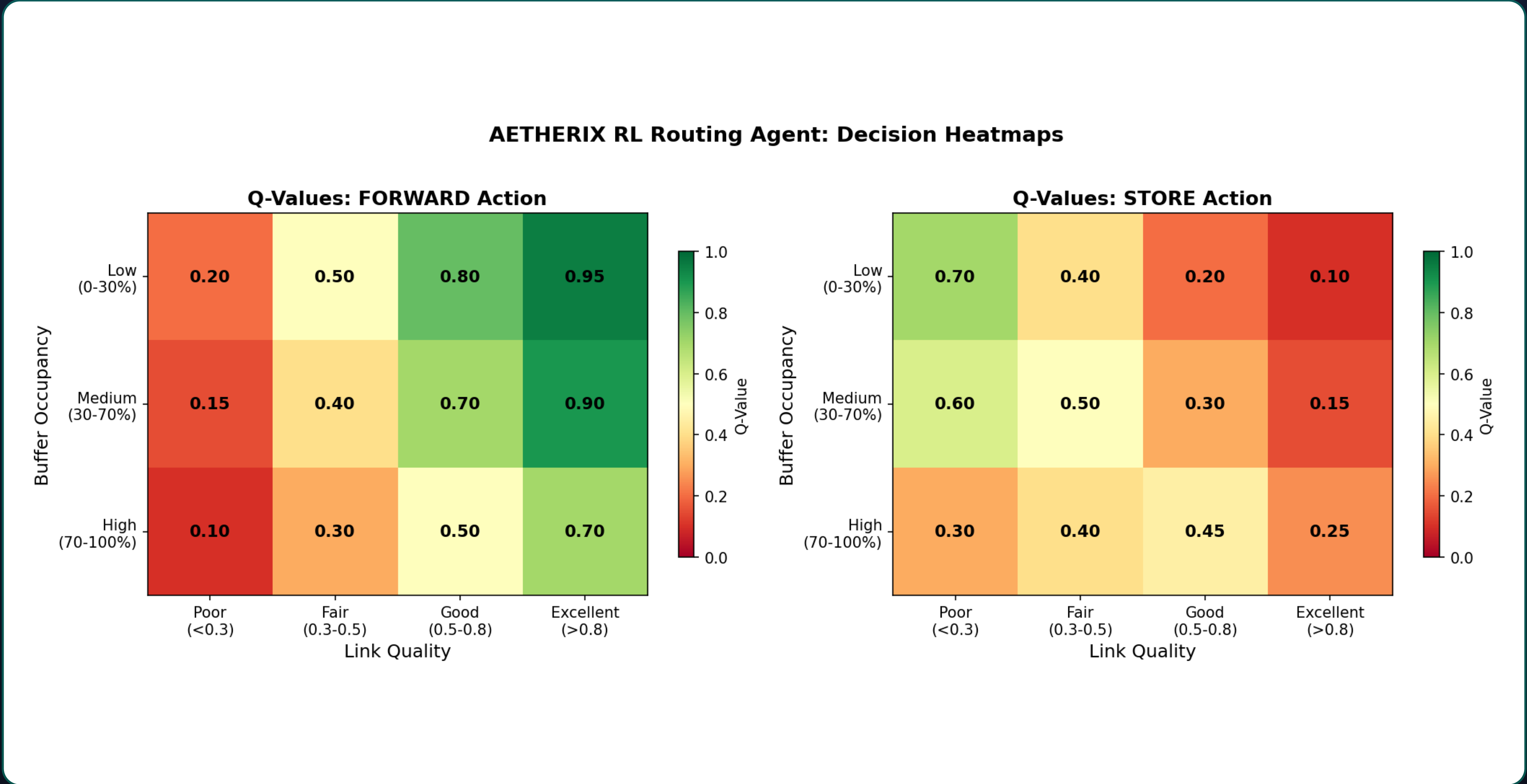
Inspectable Policy

Every Q-value human-auditable [A1]

[A1] rl_agent.py (reward fn, ε-decay 0.995) · [A3] run_simulation Module 3 (training convergence 713/800)

RL ROUTING Q-VALUE HEATMAP

Convergence of Optimal Routing Decisions



Warm colors = high-value actions · Cool colors = poor choices the agent avoids [A1]

QUANTUM SECURITY — FUTURE-PROOF

BB84/E91 QKD + Multi-Hop Repeaters + CASCADE

BB84 PROTOCOL STEPS

1. Alice sends qubits in random bases (rectilinear/diagonal)
2. Bob measures each qubit in random basis
3. Public reconciliation of basis choices (~50% retained)
4. QBER estimation: < 11% = SECURE [15], ≥ 11% = ABORT
5. CASCADE error reconciliation
6. Privacy amplification via universal hashing
7. Final shared secret key

Phase	Segment	Protocol	Range
Phase 1	Earth ⇔ Orbit	BB84 direct	36,000 km
Phase 2	Earth ⇔ Lagrange	1 repeater	1.5M km
Phase 3	Earth ⇔ Mars	3-hop chain	225M km

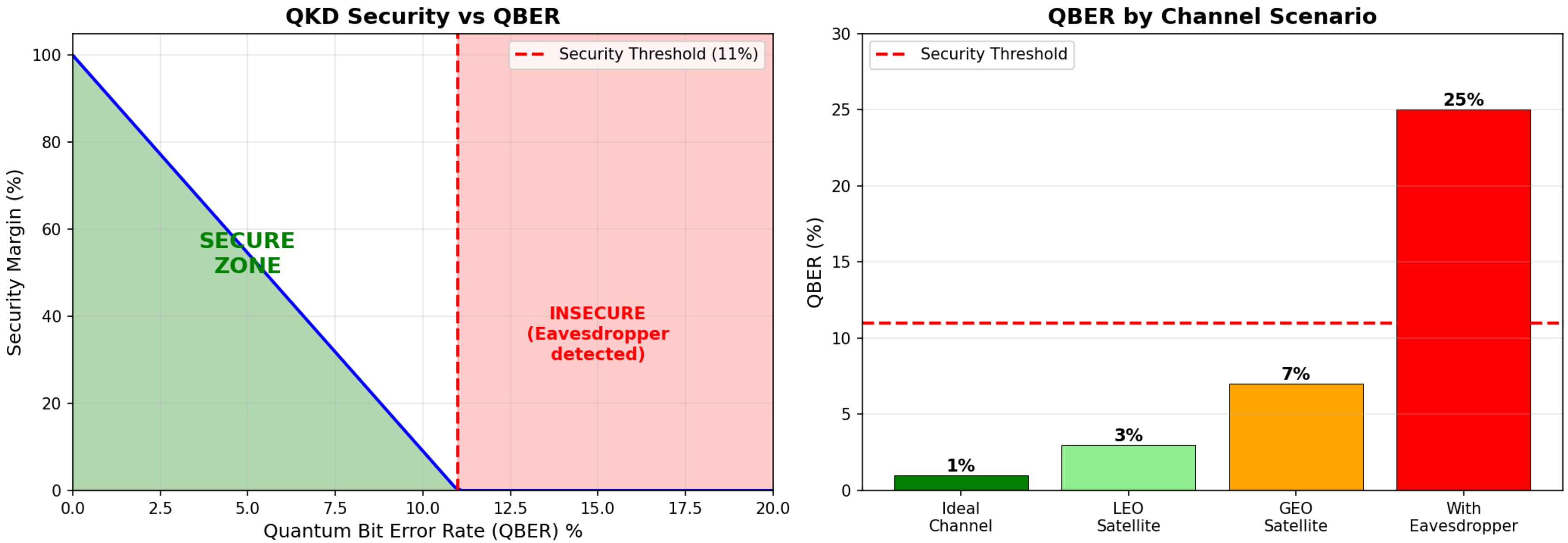
POST-QUANTUM CRYPTO (PQC)

Kyber (ML-KEM, FIPS 203) — Key encapsulation [16]
Dilithium (ML-DSA, FIPS 204) — Digital signatures [17]
Hybrid: QKD + PQC for defense in depth

[13] Bennett-Brassard 1984 BB84 · [14] Ekert 1991 E91 · [15] Shor-Preskill 2000 (QBER<11%) · [16][17] NIST FIPS 203/204 · [A5] qkd.py

QKD SECURITY ANALYSIS

QBER vs Eavesdropper Detection



QBER < 11% threshold ensures security [15] · Any eavesdropping attempt is detected [13]

ORBITAL MECHANICS & CONTACT WINDOWS

780-Day Synodic Period — From Opposition to Conjunction

Parameter	Value
Semi-major axis	1.524 AU (227.9M km)
Orbital period	686.98 days
Synodic period	779.94 days
Min distance	54.6M km (perihelion)
Max distance	401M km (aphelion)
Eccentricity	0.0934

Window	Availability	Duration	Data Rate
Optimal (opposition)	99%	8–12 hrs	100–200 Mbps
Good	95%	6–8 hrs	20–100 Mbps
Fair	85%	2–4 hrs	5–20 Mbps
Poor (aphelion)	50%	0–2 hrs	1–5 Mbps
Blackout (conjunction)	0%	N/A	Lagrange relay only

[3] JPL Horizons (orbital ephemeris: synodic 779.94 d, 54.6M–401M km) · [A2] topology.py · [A4] link_budget.py (windows)

RADIATION-HARDENED COMPUTING

Surviving the Space Radiation Environment (LO-e)

Effect	What it does	Mitigation
SEU (Single Event Upset)	Bit flip in memory/register	SECDED ECC
MBU (Multiple Bit Upset)	≥2 adjacent bits flipped	Bit interleaving
SEL (Single Event Latchup)	Parasitic short — destructive	Current limit + cycle
TID (Total Ionizing Dose)	Cumulative degradation (krad)	Rad-hard (RAD750) [18]

DEFENSE-IN-DEPTH STACK

1. TMR — 3 replicas + majority voter masks single fault
2. SECDED (39,32) ECC — corrects 1, detects 2 (21.9% OH)
3. Memory scrubbing — rewrites before 2nd upset (Poisson race)
4. FDIR + watchdog — detect → isolate → reload golden image
5. SAFE-MODE after recovery budget exhausted

DEMONSTRATED (run_simulation Module 6 [A6])

37,159
Raw upsets (210-day cruise)

200×
Protection factor

3,334×
TMR gain (@ p=1e-4)

2,127×
RAD750 TID margin

HERITAGE: NASA RAD750 (Curiosity, Perseverance C&DH) [18] • ESA LEON3FT / GR712RC [19] • Code: src/computing/radiation.py [A6]

Result above: ~186 uncorrectable residuals vs 37,159 raw upsets over a 512 Mbit interplanetary cruise — ~0.9/day residual.

MISSION-CRITICAL DATA PRIORITIZATION

Bandwidth Triage on a Starved, Intermittent Link (LO-f)

Tier	Class	Examples	BPv7 Priority
P0	Emergency / Safety	Health, collision avoidance, faults	EMERGENCY
P1	Mission-critical	Command ACKs, time-sensitive science	HIGH_SCIENCE
P2	High-priority	Routine telemetry, scheduled science	STANDARD
P4	Low / Bulk	Housekeeping, file transfers, SW images	BULK

THREE LEVERS

1. Compression (pre-transmit) [7][8]:
   - Telemetry → CCSDS 121.0-B-3 (Rice) ≈ 3× [7]
   - Imagery → CCSDS 122.0-B-2 (wavelet) ≈ 10× [8]
2. Deadline-aware preemptive QoS [A7]:
   strict priority → earliest deadline; defer if infeasible
3. BPv7 fragmentation [9]:
   send what fits this contact; defer the rest; link never idle

DEMONSTRATED SCENARIO (src/routing/prioritization.py [A7])

5/6
Items delivered (strict priority)

100%
Link utilisation (target)

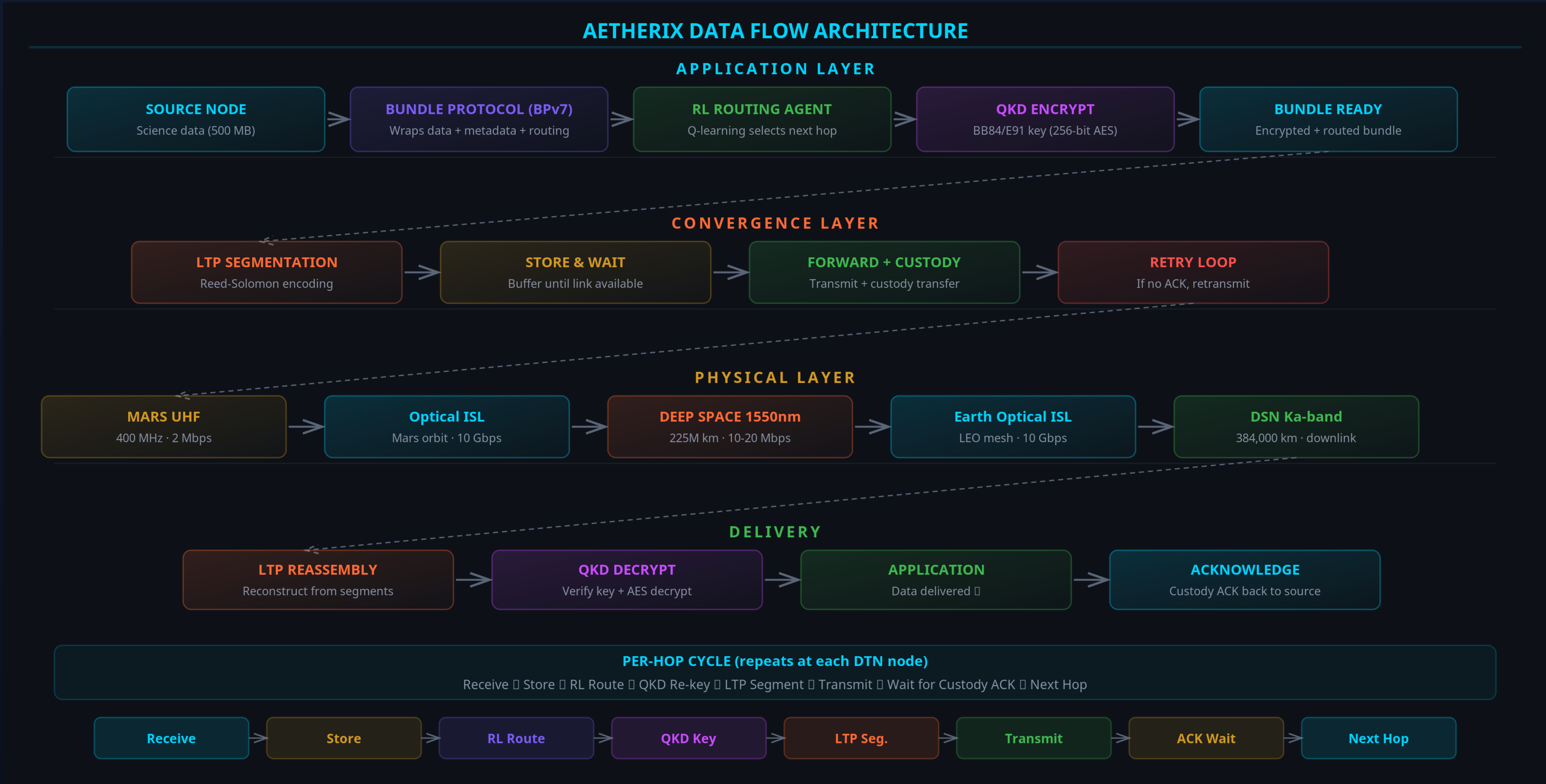
41%
SW update fragmented (remainder deferred)

P0
Emergency preempts to direct-to-Earth backup

[A7] AETHERIX prioritization.py · [7] CCSDS 121.0-B-3 lossless · [8] CCSDS 122.0-B-2 wavelet · [9] RFC 9171 (fragmentation)

DATA FLOW DIAGRAM

Complete Data Path from Mars Surface to Earth Control



PROTOCOL STACK

BPv7 + Three Convergence Layers — Standards-Compliant

AETHERIX PROTOCOL STACK

APPLICATION LAYER

Science Data	Telemetry, imagery, spectral analysis
Commands	Uplink instructions, configuration
Telemetry	Spacecraft health, navigation data

BUNDLE PROTOCOL v7 (RFC 9171) — RFC 9171 / CCSDS 734.2-B-1

P0 EMERGENCY	Life-critical, immediate delivery
P1-2 PRIORITY	Command & control, high-value science
P3 NORMAL	Routine data, standard telemetry
P4 BULK	Archive data, software updates

CONVERGENCE LAYER — RFC 5326 / RFC 7242

LTP (RFC 5326)	TCPCL (RFC 7242)	UDP-CL
Deep space links	Earth ground segment	Optical ISL
Reliable, retransmission	Session-based	Low-latency, fragmented

SECURITY LAYER — NIST FIPS 203 / FIPS 197

QKD: BB84/E91	Quantum key distribution
Post-Quantum: ML-KEM	NIST FIPS 203 lattice-based
AES-256-GCM	Symmetric encryption

PHYSICAL LAYER — CCSDS 141.0-B-1

UHF 400 MHz	Mars surface, short range
X-band 8.4 GHz	Near-Earth telemetry
Ka-band 32 GHz	High-rate downlink
1550nm Optical	Deep space, 10-200 Mbps

AETHERIX vs CURRENT SYSTEMS

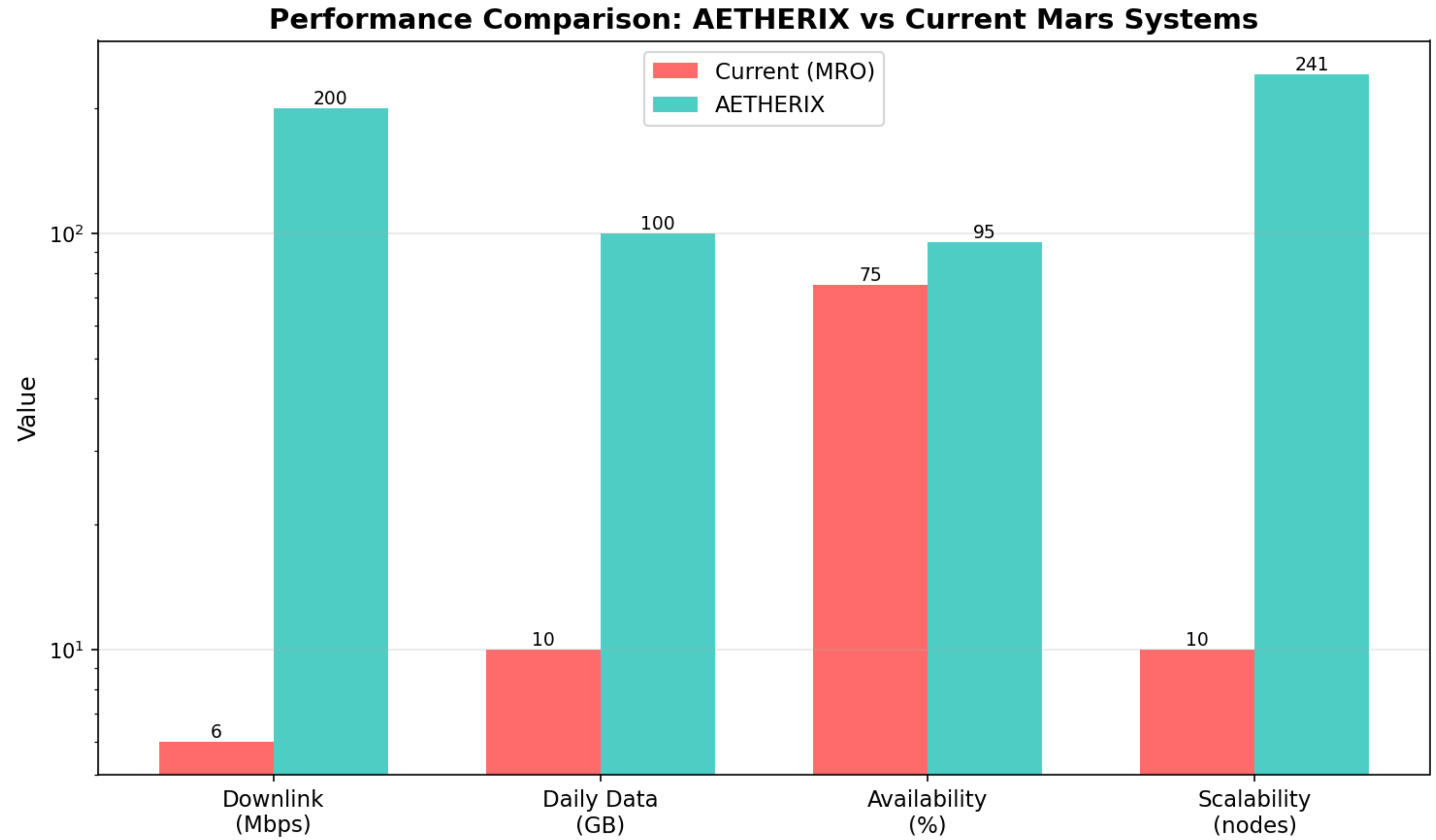
Head-to-Head Performance Comparison

Metric	Current (MRO) [1]	AETHERIX [A4]	Note
Downlink rate	0.5–6 Mbps	2–200 Mbps	10–100× capability
Daily volume	5–10 GB	50–100 GB	Est. (link-driven)
Availability	60–75%	>95%	Design target
Routing	Static (CGR)	RL-adaptive	Adaptive [A1]
Security	AES-256	QKD + PQC	Future-proof [16][17]
Scalability	5–10 assets	241 nodes	10–100× [A2]
Conjunction	Blackout	50–70% via L4/L5	+50–70% [A8]
Recovery time	Manual (hours)	Auto (seconds)	Auto [A1]

[1] NASA MRO (0.5–6 Mbps) · [A4] link_budget.py (2–200 Mbps capability) · [A2] topology.py · [A8] Module 4 conjunction · targets clearly labelled

AETHERIX vs CURRENT SYSTEMS

Head-to-Head Performance Metrics



10–100× data-rate capability (link budget [A4] vs [1]) · availability/routing are design targets

TRADE-OFF ANALYSIS — WHY THESE CHOICES

Every Decision Traded Performance for Auditability & Reproducibility

Decision	Choice	Rationale (vs alternative)
Optical vs RF	Hybrid: 1550 nm primary + Ka-band fallback [4]	10–100× throughput [A4]; RF survives clouds & corona
Routing	Custom Q-learning, not ION-DTN CGR [A1]	Adapts to live state; CGR re-plans on 12-min-stale schedule
RL model	Q-tables now, DQN later (Phase 6)	Every Q-value human-auditable; trains in seconds
State space	Discretised, 241 nodes [A2]	Right-sized for a tabular policy; DQN path documented
Reward weights	α=1.0, δ=10.0, ε-decay 0.995 [A1]	Drop penalty 10× delivery to forbid bundle loss

BOTTOM LINE

Each choice sacrifices maximum theoretical performance for auditability and reproducibility — exactly what a defence and a research artefact require. DQN / ns-3 / ION-DTN are the documented production transition (Phases 7–9).

DSOC heritage: NASA flew optical + RF side-by-side on Psyche [4] — AETHERIX mirrors that hybrid model.

[A1] rl_agent.py / training.py · [A4] link_budget.py · [4] NASA DSOC (Psyche) · [A2] topology.py

FAILURE & RECOVERY

Autonomous Solar-Conjunction Link-Blackout Survival

SCENARIO: Earth–Sun–Mars conjunction — solar corona collapses the 1550 nm link below the 0.3 forward threshold [A1]

Path	Band	Link Quality	Status	Reward [A8]
			SED	
			EN	
			EN	

HOW AETHERIX RECOVERS (automatically)

1. Detect — optical Q-value collapses (q<0.3 → no reward)
2. Re-route — agent in exploit mode picks highest-Q:
   ES-L4 (Ka-band RF), 60° solar elongation, avoids corona
3. Prioritise — policy engine fires two rules:
   P0 EMERGENCY → forward on best (Ka-band) link
   P4 BULK → store locally, defer past conjunction

WHY LAGRANGE RELAYS [3][A2]

ES-L4 / ES-L5 sit 60° ahead/behind Earth in its orbit.
At that separation they keep line-of-sight to Mars
around the solar limb even at true conjunction.
Direct Earth–Mars link: 0% availability at conjunction.
Via ES-L4/L5: 50–70% availability retained [A8].
No Mars-orbit relay can solve this — geometry is Earth-side.

Outcome: throughput drops (optical→RF) but no mission-critical data lost. Run live: python run_simulation.py --module 4

[A8] run_simulation Module 4 (rewards −1.438 / −0.201) · [A1] rl_agent.py (0.3 threshold) · [3] JPL Horizons (Lagrange geometry) · [A2] topology.py

IMPLEMENTATION

25 Modules, 480 Tests, Full Python Stack

25
Source Modules

480
Unit Tests

241
Network Nodes

12
Interactive Demos

CORE MODULES

- link_budget.py — Optical/RF link analysis
- rl_agent.py — Q-learning routing agent
- bundle.py — BPv7 bundle protocol
- forwarding_engine.py — Store-and-forward
- qkd.py — BB84/E91 protocols
- contact_windows.py — Orbital mechanics
- topology.py — 5-tier network (241 nodes)
- simulator.py — End-to-end simulation
- ltp.py — Licklider Transmission Protocol
- tcpcl.py — TCP Convergence Layer

Standard	Title	Status
RFC 9171 [9]	Bundle Protocol v7	Compliant
RFC 4838 [12]	DTN Architecture	Compliant
CCSDS 734.2-B-1 [2]	CCSDS Bundle Protocol	Compliant
CCSDS 734.3-B-1	SABR / contact routing	Baseline
CCSDS 141.0-B-1 [5]	Optical Comm Physical	Compliant
RFC 9172	BPSec (security)	Referenced
RFC 5326 [10]	LTP (deep space)	Compliant
RFC 7242 [11]	TCPCL (Earth)	Compliant

[9][10][11][12] IETF RFCs · [2][5] CCSDS standards · [A1]–[A8] AETHERIX modules (github.com/matx104/AETHERIX)

ROADMAP & FUTURE WORK

From Proof-of-Concept to Production Deployment

Phase 1–2: Foundation
Complete

Topology + BPv7 + Convergence Layers

Phase 3: Intelligence
Complete

RL Routing (Federated Q-learning)

Phase 4: Security
Complete

QKD + Multi-hop Repeaters + CASCADE

Phase 5–6: Simulation
Complete

Full Engine + Web Platform + Demos

Phase	Focus	Technology	Status
7	DQN Upgrade	Deep Q-Network	Remaining
8	HIL Simulation	ns-3 integration	Remaining
9	ION-DTN	Production DTN stack	Remaining
10	Flight Demo	LEO optical test	Future
11	Mars Deploy	Full 5-tier network	Future

[A1]–[A8] AETHERIX source modules · [4] NASA DSOC (Psyche) heritage for optical · DQN/ns-3/ION-DTN on production roadmap

CONCLUSION — AETHERIX DELIVERS

Bridging the Interplanetary Communication Gap

THE PROBLEM

- 54.6–401M km Earth-Mars distance [3]
- 3–22 min one-way light delay [3]
- TCP/IP fundamentally broken in space [12]
- Current RF: only 0.5–6 Mbps [1]
- 2-week solar conjunction blackouts [3]

THE SOLUTION

- BPv7 DTN: store-and-forward works [9]
- RL routing: adaptive, auditable policy [A1]
- Optical: 10–100× data-rate capability [A4]
- QKD: future-proof quantum security [13][15]
- Lagrange relays: conjunction coverage [A8]

10–100×
Target Speedup

>95%
Availability (target)

AI-Driven
Routing

Quantum
Security

NOVEL CONTRIBUTIONS

RL autonomous routing (auditable Q-table) [A1] | 5-tier DTN (241 nodes) [A2] | Optical/RF hybrid (10-100x capability) [A4] | Quantum links (future-proof) [A5] | Full simulatior

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[20] IETF, RFC 4838 §3.1, 2007 (long delay, intermittent connectivity challenges).

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Project Sources [A1]–[A8] · M. A. Tariq, AETHERIX, github.com/matx104/AETHERIX (2025)

[A1] RL Routing Agent — src/routing/rl_agent.py

[A2] Network Topology — src/orbital/topology.py

[A3] End-to-End Simulation — run_simulation.py Module 3 (RL training)

[A5] QKD Protocol Implementation — src/security/qkd.py

[A6] Radiation Hardening Model — src/computing/radiation.py

[A7] Data Prioritization Engine — src/routing/prioritization.py

THREE-LAYER ATTRIBUTION (resolves the prior citation gap)

Layer A — Industry/Scientific baseline [1]–[20]: someone else's data (e.g. NASA MRO 0.5–6 Mbps [1]).

Layer B — This project's design decision [A1]–[A8]: cite the code (e.g. RL agent src/routing/rl_agent.py [A1]).

Layer C — Demonstrated simulation results [A6][A8]: cite the specific run (e.g. radiation 200× — Module 6 [A6]).

Design targets (e.g. >95% availability, 2–200 Mbps capability) are labelled as such, never presented as measured results.

THANK YOU

Questions?

10–100×	>95%	241	480
Faster	Available (target)	Nodes	Tests

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LIVE DEMO & RESOURCES

matx104.github.io/AETHERIX — Interactive simulations

github.com/matx104/AETHERIX — Source code + documentation

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